

Quantum Mechanics Master Formula Sheet

de Broglie Waves • Wave Equations • Operators • Potential Well • Uncertainty Principle • Tunnelling

1. de Broglie Wave Hypothesis & Matter Waves

Fundamental Wavelength: $\lambda = h/p = h/(mv) = h/\sqrt{2mE}$

Special Acceleration & Thermal Cases

• **Electron Accelerated by Potential $\$V\$$:**

$$\lambda = h/\sqrt{2m_e V} = 1.226/\sqrt{V} \text{ nm} = 12.26/\sqrt{V} \text{ \AA}$$

• **Neutrons / Gas Molecules at Temperature $\$T\$$:**

$$\lambda = h/\sqrt{3m k_B T}$$

• **Relativistic Particle Wavelength:**

$$\lambda = h/(\gamma m_0 v) = (h/m_0 v) \cdot \sqrt{1 - v^2/c^2}$$

2. Phase Velocity (v_p) & Group Velocity (v_g)

Phase Velocity (Wave Speed): $v_p = \omega/k = f\lambda = E/p$

Group Velocity (Particle Speed): $v_g = d\omega/dk = dE/dp = v_{\text{particle}}$

Key Relations

• **Non-Relativistic Free Particle:** $v_p = \frac{1}{2} v_g$

• **Relativistic Particle Relation:** $v_p \cdot v_g = c^2$

• **Dispersive Media Formula:** $v_g = v_p - \lambda (dv_p/d\lambda)$

3. Heisenberg Uncertainty Principle

Position & Momentum: $\Delta x \cdot \Delta p_x \geq \hbar/2 = h/(4\pi)$

Energy & Time: $\Delta E \cdot \Delta t \geq \hbar/2$

Angular Position & Momentum: $\Delta \theta \cdot \Delta L \geq \hbar/2$

Major Applications

• **Non-existence of electron in nucleus:** Required $E_k \approx 20 \text{ MeV}$ vs max observed $E \approx 4 \text{ MeV}$.

• **Spectral Line Width:** $\Delta E = \hbar/(2\Delta t)$ (where Δt is excited state lifetime).

• **Hydrogen Atom Ground Energy:** $E_0 = -m_e e^4 / (32 \pi^2 \epsilon_0^2 \hbar^2) \approx -13.6 \text{ eV}$.

4. Wave Function (Ψ) & Probability Properties

1D Free Particle Wave Function: $\Psi(x, t) = A e^{i(kx - \omega t)} = A [\cos(kx - \omega t) + i \sin(kx - \omega t)]$

• **Born's Probability Density:** $P(x, t) = |\Psi(x, t)|^2 = \Psi^*(x, t) \cdot \Psi(x, t)$

• **Normalization Condition:** $\int_{-\infty}^{+\infty} |\Psi(x, t)|^2 dx = 1$

• **Probability in Interval $[x_1, x_2]$:** $P = \int_{x_1}^{x_2} |\Psi(x, t)|^2 dx$

Expectation Value Formulas

• **Position:** $\langle x \rangle = \int_{-\infty}^{+\infty} \Psi^* x \Psi dx$

• **Momentum:** $\langle p_x \rangle = \int_{-\infty}^{+\infty} \Psi^* (-i\hbar \partial/\partial x) \Psi dx$

• **Energy:** $\langle E \rangle = \int_{-\infty}^{+\infty} \Psi^* (i\hbar \partial/\partial t) \Psi dx$

5. Quantum Mechanical Operators

Physical Quantity	Operator Symbol	Mathematical Operator Form
Position	$\hat{x}$	x
Linear Momentum	$\hat{p}_x$	$-i\hbar (\partial / \partial x)$
Kinetic Energy	T	$-(\hbar^2 / 2m) (\partial^2 / \partial x^2)$
Potential Energy	V	$V(x)$
Hamiltonian (Total Energy)	H	$-(\hbar^2 / 2m) (\partial^2 / \partial x^2) + V(x)$
Energy Operator	E	$i\hbar (\partial / \partial t)$

6. Schrödinger Wave Equations

Time-Dependent Schrödinger Equation (1D)

$$i\hbar (\partial \Psi / \partial t) = -(\hbar^2 / 2m) (\partial^2 \Psi / \partial x^2) + V(x) \Psi$$

Time-Independent Schrödinger Equation (1D)

$$(d^2 \psi / dx^2) + (2m / \hbar^2) [E - V(x)] \psi = 0$$

where wave number $k = \sqrt{2m(E - V) / \hbar^2} \Rightarrow (d^2 \psi / dx^2) + k^2 \psi = 0$.

7. Eigenvalues & Eigenfunctions

General Operator Eigenvalue Eq: $\hat{A} \psi_n = a_n \psi_n$

Hamiltonian Eigenvalue Eq: $H \psi_n = E_n \psi_n$

*Where $\hat{A}$ is a linear operator, ψ_n is the eigenfunction, and a_n is the corresponding physical eigenvalue scalar.

8. Particle in a 1D Infinite Potential Well (Box Width L)

Potential Profile: $V(x) = 0$ for $0 \leq x \leq L$; and $V(x) = \infty$ elsewhere.

Energy Eigenvalues: $E_n = (n^2 \hbar^2) / (8m L^2) = (n^2 \pi^2 \hbar^2) / (2m L^2)$ ($n = 1, 2, 3, \dots$)

- **Ground State Energy (\$n=1\$):** $E_1 = \hbar^2 / (8m L^2)$
- **Excited State Energies (\$n > 1\$):** $E_n = n^2 E_1$

Normalized Wave Functions: $\psi_n(x) = \sqrt{2/L} \cdot \sin(n\pi x / L)$ ($n = 1, 2, 3, \dots$)

Probability Density Nodes & Maxima

- **Ground State (\$n=1\$):** Peak probability density at $x = L/2$ with $P_{max} = 2/L$.
- **1st Excited State (\$n=2\$):** Peaks at $x = L/4, 3L/4$; zero probability node at $x = L/2$.

9. Quantum Tunnelling & Potential Barrier

Barrier Profile: $V(x) = V_0$ for $0 \leq x \leq a$, and $V(x) = 0$ elsewhere.

Transmission Coefficient (\$T\$) for \$E < V_0\$: $T \approx e^{-2\alpha a}$

where attenuation constant $\alpha = \sqrt{2m(V_0 - E) / \hbar^2} = (2\pi / h) \sqrt{2m(V_0 - E)}$.

Conservation Law: $R + T = 1$ (Reflection + Transmission = 1)

10. Fundamental Physical Constants

Constant Name	Symbol	Value with Standard Units
Planck's Constant	h	$6.626 \times 10^{-34} \text{ J}\cdot\text{s}$
Reduced Planck's Constant	$\hbar = h / (2\pi)$	$1.054 \times 10^{-34} \text{ J}\cdot\text{s}$
Electron Rest Mass	m_e	$9.109 \times 10^{-31} \text{ kg}$
Proton Rest Mass	m_p	$1.672 \times 10^{-27} \text{ kg}$
Neutron Rest Mass	m_n	$1.675 \times 10^{-27} \text{ kg}$
Elementary Charge	e	$1.602 \times 10^{-19} \text{ C}$
Speed of Light in Vacuum	c	$3.00 \times 10^8 \text{ m/s}$
Boltzmann Constant	k_B	$1.380 \times 10^{-23} \text{ J/K}$
Electron-Volt Conversion	1 eV	$1.602 \times 10^{-19} \text{ J}$